

Threat Modeling Diagram and Explanation

This document explains the threat modeling diagrams used in the Secure Cloud Banking Application project on AWS. These diagrams help to understand system architecture, possible cyber attacks, security controls, and the incident response process.

Diagram 1 - Secure Cloud Banking Architecture

This diagram shows the overall AWS cloud architecture of the secure banking application. It explains how users connect to the application through the internet and how requests are protected using AWS WAF and AWS Shield. The architecture contains Web Servers, Application Servers, Amazon RDS database, Amazon S3 storage, IAM, KMS, and monitoring services like CloudWatch and CloudTrail. This diagram mainly represents the secure deployment and communication flow inside the AWS cloud environment.

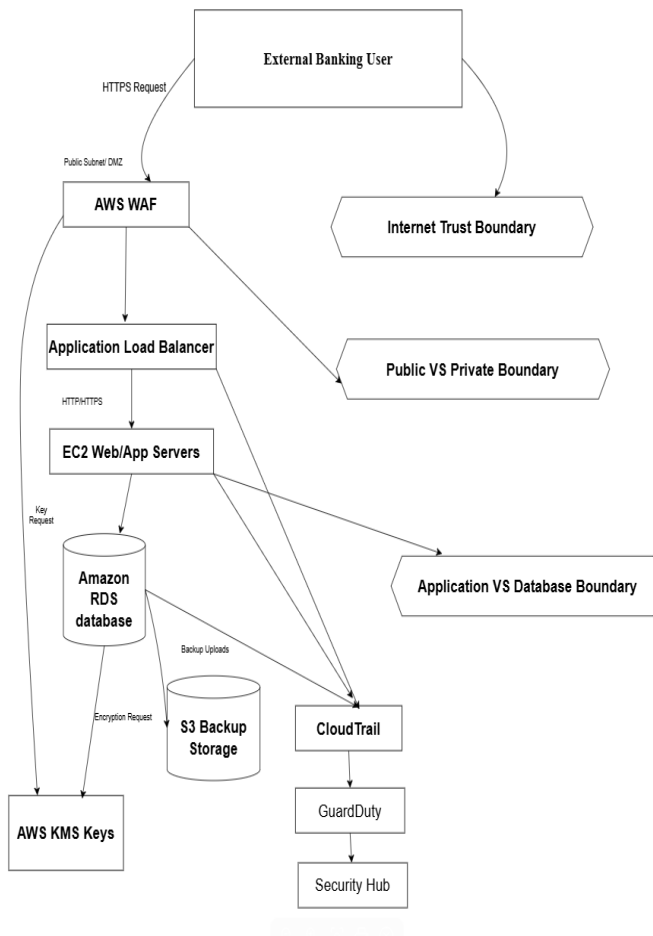


Figure 1- Secure Cloud Banking Architecture

Diagram 2 - Threat and Attack Flow

This diagram explains the possible cyber attacks that can occur in the cloud environment. It highlights attacks such as SQL Injection, Brute Force attacks, DDoS attacks, Malware, Ransomware, and Unauthorized Access. The red color arrows represent attack paths and attack targets. The diagram also shows how AWS security services detect and block these threats using WAF, GuardDuty, Security Hub, and Security Groups.

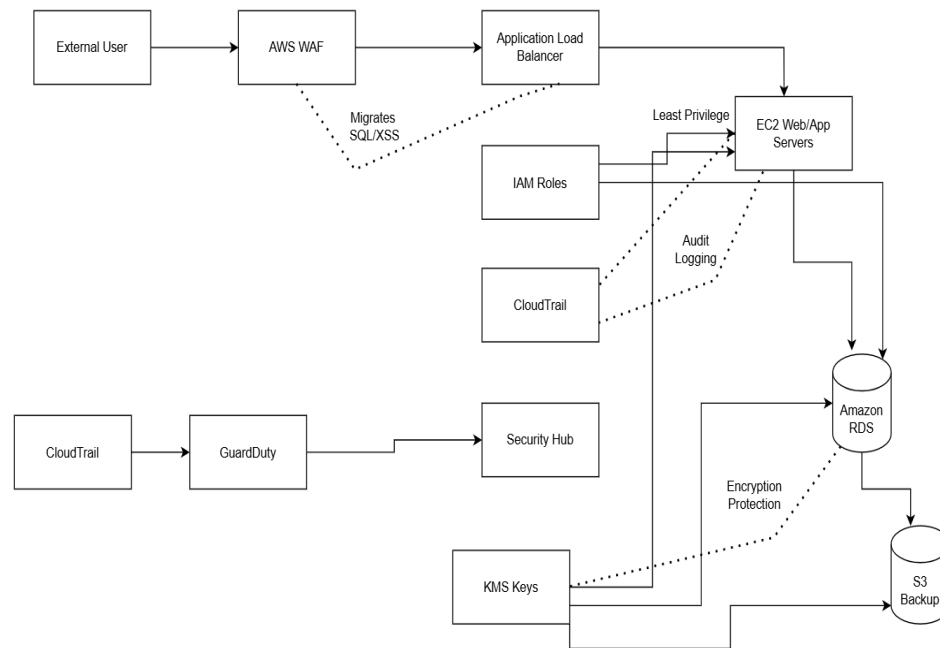


Figure 2- Threat and Attack Flow

Diagram 3 - Incident Response Process

This diagram represents how security incidents are handled in the cloud banking application. When a threat is detected, monitoring tools generate alerts. The SOC team investigates the issue, classifies the severity level, and performs containment actions such as isolating EC2 instances, blocking IPs, or disabling IAM users. After containment, eradication and recovery steps are performed to restore the application safely. Finally, a Root Cause Analysis (RCA) and lessons learned process is conducted.

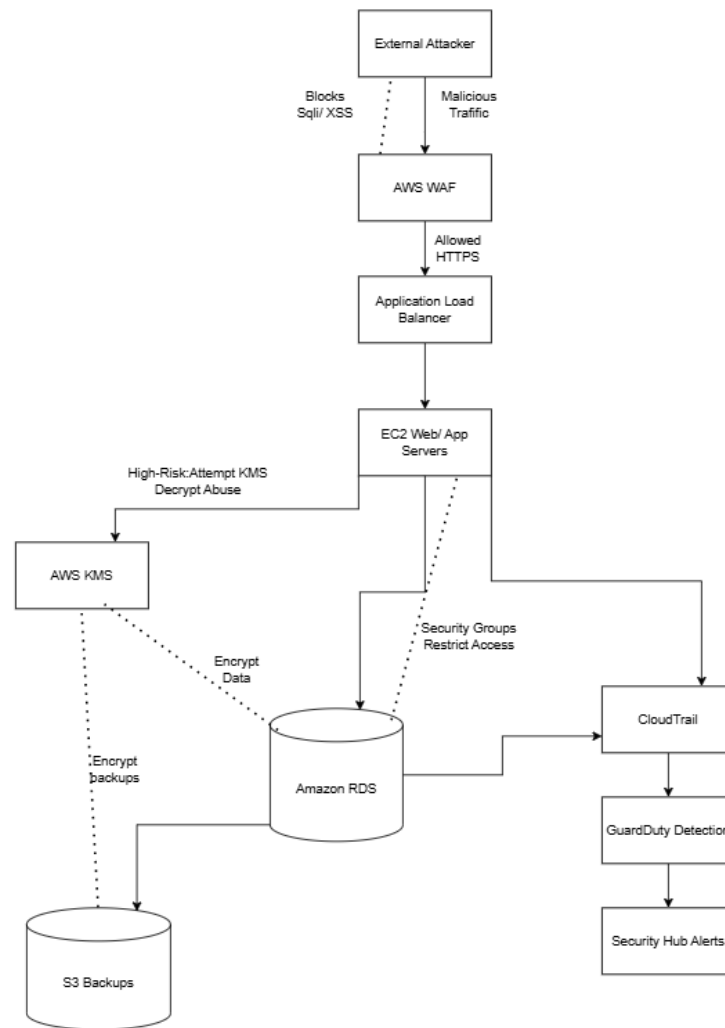


Figure 3 - Incident Response Process

Conclusion

The threat modeling diagrams provide a clear understanding of the secure cloud banking system, possible attack scenarios, and the incident handling process. These diagrams help in identifying vulnerabilities, improving cloud security, and implementing proper monitoring and response mechanisms in AWS.